

Introduction to Bayesian estimation techniques

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The basic framework

Linear models

- We observe certain quantities, and we know that there are relationships between the observed values
- We use models to describe these relationships
- Probably the simplest model is a linear relationship between two variables y and x

$$y_t = \mu + \beta x_t \tag{1}$$

- We observe several corresponding values $(y_1, x_1), (y_2, x_2), \dots, (y_T, x_T)$
- If the relationship really truly holds without error, then there's not much to do: any two pairs of observations will determine the value μ and β

Stochastic models

- In realistic circumstances, such exact relationships are very unlikely (or there's no unknown parameter to be estimated)
- A non-exact relationship is one where there are discrepancies

$$y_t = \mu + \beta x_t + \varepsilon_t \quad (2)$$

- We still observe $(y_1, x_1), (y_2, x_2), \dots, (y_T, x_T)$
- Without further restrictions on ε_t , this model is pretty useless: for any values of μ and β , there's a corresponding series $\varepsilon_1, \dots, \varepsilon_T$ such that the model equation holds

Error term properties

- So we make assumption about ε_t
- If the model is a reasonable description of the actual reality, then we can expect that
 - On average, there's no consistent error
 - On average, the discrepancies are not arbitrarily large in magnitude

- The first condition leads to the mathematical formulation

$$\frac{1}{T} \sum_t \varepsilon_t \approx 0, \quad \text{for large } T \quad (3)$$

- The mathematical formulation of the second condition is not as obvious; a common form is

$$\frac{1}{T} \sum_t \varepsilon_t^2 \approx \sigma^2, \quad \text{for large } T \quad (4)$$

with some value σ^2

Error term standard deviations

- We have added a new parameter σ , which we call the *error term standard deviation*
- But adding this new parameter does not solve the problem: if arbitrary large σ is possible, the model is still useless
- Two solutions
 - require the *sample standard deviation* $\frac{1}{T} \sum_t \varepsilon_t^2$ to be as small as possible
 - make further assumptions about the *distribution* of the error term, enabling us to assess the *likelihood* of the observed sample
- The first leads to (ordinary) least squares (OLS), the second to maximum likelihood (ML)

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The classical approach

OLS estimates

- The OLS estimate of μ , β are the values which minimize the *residual sum of squares*

$$\sum_t (y_t - \mu - \beta x_t)^2 \quad (5)$$

- The solution is

$$\hat{\beta} = \frac{\sum_t x_t y_t - (\sum_t x_t)(\sum_t y_t)}{\sum_t x_t^2 - (\sum_t x_t)^2} \quad (6)$$

$$\hat{\mu} = \sum y_t - \hat{\beta} \sum_t x_t \quad (7)$$

OLS inference

- We usually also say that

$$\hat{\sigma}^2 = \frac{1}{T} \sum_t (y_t - \hat{\mu} - \hat{\beta}x_t)^2 \quad (8)$$

- So we have our *point estimates*, but what have we learnt about the *true values* of μ , β and σ^2 ?
- We know how to calculate (usually asymptotically valid) *confidence intervals*, using the point estimates
- We usually say that these confidence intervals contain the true values with a given probability

Classical confidence intervals

- But we also know that this is an incorrect interpretation of the confidence interval
- The conceptual problem is that in the classical approach, the true value is a fixed number, and not a random variable
- Therefore it makes no sense to talk about the probability of the true value being in some interval
- In the classical approach, the sample is random, and therefore the point estimates are also random
- Hence the correct interpretation is that the *random confidence interval*, calculated for a large number of *simulated samples*, will contain the true value with the given probability

Maximum likelihood estimation I

- Defining good fit as minimized sum of squares is a somewhat ad hoc choice
- Why not the sum of absolute values? Or the sum of the fourth powers?
- If we want to be less ad hoc, we have to make additional assumptions
- Which will hopefully be not very ad hoc ...

Maximum likelihood estimation II

- Let's assume the we can characterize the probability distribution of the discrepancies as a function of the model parameters
- Then it is reasonable to search for those parameter values which maximize this probability
- In real life cases we most often work with continuous distributions, so that each sample has probability 0
- We use the probability distribution function to work with the *likelihood* instead of he probability

Normally distributed error terms

- Let's return to the example of the linear model

$$y_t = \mu + \beta x_t + \varepsilon_t \quad (9)$$

- Assume that the error terms are normally distributed

$$\varepsilon_t \sim \mathcal{N}(0, \sigma^2) \quad (10)$$

- In other words, the PDF is ε_t is

$$f(z) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{z^2}{2\sigma^2}} \quad (11)$$

Normal likelihood

- What will then the likelihood of the sample be?
- We need also to describe the distribution of the x_t -s
- And also the relationship between ε_t and ε_s for $t \neq s$
- The standard assumptions are
 - the error terms are independent
 - x_t -s are in fact not random, but fixed values
- The latter assumption again corresponds to the classical case when the data comes from controlled and repeatable experiments, and implies that $y_t \sim \mathcal{N}(\mu + \beta x_t, \sigma^2)$
- The likelihood of the sample will then be

$$\mathcal{L}(Y_T; \mu, \beta, \sigma^2) = \prod_{t=1}^T \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_t - \mu - \beta x_t)^2}{2\sigma^2}} \quad (12)$$

Normal ML estimated

- Maximizing the product is rather cumbersome, so we turn it into a sum by taking the logarithm, which does not change the location of the maximum

$$\begin{aligned}\ell(Y_T; \mu, \beta, \sigma^2) &= \sum_{t=1}^T -\log(\sqrt{2\pi\sigma^2}) - \sum_{t=1}^T \frac{(y_t - \mu - \beta x_t)^2}{2\sigma^2} = \\ &= -\frac{T}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=1}^T (y_t - \mu - \beta x_t)^2\end{aligned}\tag{13}$$

- Since maximizing the log-likelihood is equivalent to minimizing the sum of squares, the normal ML estimates are equal to what OLS would give
- This need not be the case for non-normal error distributions

Main features of the classical method

- Even though the final formulas might be different, ML estimation is still a classical method
 - It assumes that there are fixed "true values" of the parameters
 - It produces confidence intervals, determined fully by the observed data
 - Probability statements about the true values are actually statements about the confidence interval
- The classical approach was designed to be useful when drawing conclusions from controlled experiments
- But in when working with *time series data*, re-sampling is hardly an option
- This is especially true when the data is generated by a complex system, such as the economy

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The Bayesian approach

A different worldview

- Intuitively, we do statistical analysis to improve our knowledge about the world
- We might have had some knowledge *before observing the data*, and observing the data *increases the precision of this knowledge*
- The classical approach can not formalize this intuition
 - While treating hard restrictions on parameters is possible, soft *a priori* knowledge can not really be incorporated consistently
 - The classical confidence interval is conceptually not a good representation of *a posteriori* precision
- At the heart of all these issues is the assumption that true fixed parameter values exist

Prior and posterior distribution

- Everything is solved if we instead are willing to treat the model parameters as random variables
- The estimation modifies the distribution: possibly shifts the location and increases the precision
- The knowledge before observing the data is represented by the *prior distribution*
- After the data we have the *posteriori distribution*
- The key step is how we go from one to the other
- Bayes' Theorem combines the prior distribution and the likelihood to get the posterior distribution

Bayes' Theorem: discrete case

- Start with the definition of conditional probability

$$P(A | B) = \frac{P(AB)}{P(B)}, \text{ if } P(B) \neq 0 \quad (14)$$

- Let's write the same equation, but with A and B reversed

$$P(B | A) = \frac{P(AB)}{P(A)}, \text{ if } P(A) \neq 0 \quad (15)$$

- We have two expressions for the joint probability $P(AB)$

$$P(AB) = P(A | B) P(B) = P(B | A) P(A) \quad (16)$$

- And from here we get Bayes' Theorem

$$P(B | A) = \frac{P(A | B) P(B)}{P(A)} \quad (17)$$

Interpretation of Bayes' Theorem

- We think of B as a property we want to investigate, like $a < \beta < b$
- The $P(B)$ represents our initial guess of the probability of this "event"
- Then we observe the data, represented by A , which is a function of β
- The connection between A and B is such that we can evaluate $P(A | B)$, the probability of observing A given B ; this is the likelihood
- And now Bayes' Theorem tells us how to calculate the probability of B given the observation A
 - *Multiply the prior and the likelihood, and scale it so that it's a probability*

Bayes' Theorem: the continuous case

- Since we usually work with continuous distributions and not discrete events, we need a generalization of the Bayes' Theorem
- This is it

$$f(\theta|y) = \frac{f(y|\theta)f(\theta)}{f(y)} = \frac{f(y|\theta)f(\theta)}{\int f(y|\theta)f(\theta)d\theta} \quad (18)$$

- In most cases the normalizing constant is not relevant to us, so we write

$$f(\theta|y) \propto f(y|\theta)f(\theta) \quad (19)$$

meaning the posterior is *proportional to* the product of the prior and the likelihood

Bayesian point estimates

- Bayesian methods are all about distributions
- But in practice we still often want a point estimate
- This can be any "location" statistics of the posterior distribution: the mean, the mode, or even the median
- In most cases, the posterior mode is the easiest to calculate, as it can be done without calculating the normalizing constant
- On the other hand, the numerical optimization algorithm can often be hard to handle

Monte-Carlo methods

- Unfortunately, the PDF of the posterior in most cases can not be determined analytically
- Fortunately, there are methods to simulate from the posterior distribution, even if the distribution is known only up to a scale factor
- Once we have a sample, we can calculate everything we need
- Just need to keep in mind that the results will contain sampling uncertainty

Bayesian coverage intervals

- Bayesian coverage intervals correspond to the classical confidence interval
- In this case, however, there's no conceptual issue with probability statements about non-random quantities
- They can be calculated either analytically, or from a Monte-Carlo sample
- In the first case, and with a particular set of priors, *Bayesian coverage intervals coincide with classical confidence intervals*
- In these cases, the main value added of the Bayesian approach is "only" the conceptual clarity

Bayesian estimation in practice

- In practice, and especially in econometrics, the main value added of the Bayesian approach is the ability to incorporate soft prior knowledge consistently
- The econometrician task is therefore to choose the reasonable prior distribution
- The form is in most cases straightforward; the hard part is to come up with prior parameters
 - the prior mean, which is a reasonable initial guess
 - the prior precision (inverse variance), which represents uncertainty of the initial guess
- The greatest temptation to avoid is to set large prior precision, and still call the result an “estimate representing the data”

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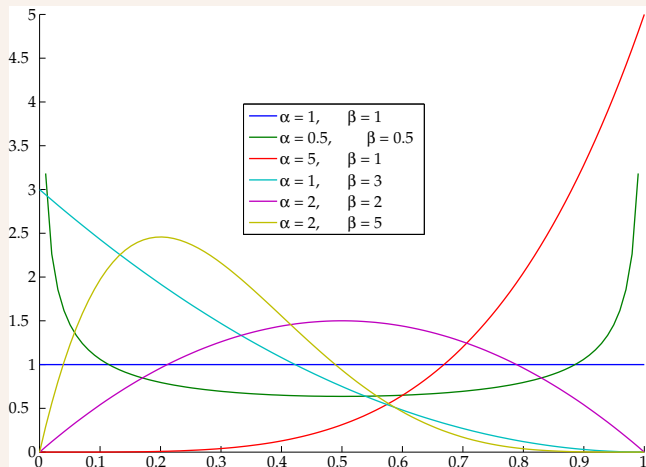
Examples

Coin flipping

- We're flipping an uneven coin, which results in a head with probability p , and we want to estimate p
- Given n observations, the likelihood is easy: it will be $f(y|p) = p^k(1 - p)^{n-k}$, where k is number of heads
- The maximum likelihood estimate is $\frac{k}{n}$
- We need to formulate a prior distribution
- In principle it can be any probability distribution which does not assign nonzero probability to "impossible" parameter values
 - So in this case, it should be concentrated between 0 and 1
- But it is often convenient (and reasonable) to use a form similar to the likelihood

β -distribution

$$f(p) = Cp^{\alpha-1}(1-p)^{\beta-1}, \quad 0 \leq p \leq 1, \quad 0 < \alpha, \beta$$



Coin flipping (cont.)

- Let the prior be a *beta distribution* with parameters α and β

$$f(p) = Cp^{\alpha-1}(1-p)^{\beta-1}, \quad 0 \leq p \leq 1, \quad 0 < \alpha, \beta$$
$$C = \frac{1}{\int_0^1 p^{\alpha-1}(1-p)^{\beta-1} dp} := \frac{1}{B(\alpha, \beta)} \quad (20)$$

- The posterior is then

$$\begin{aligned} f(p|y) &= \frac{[p^k(1-p)^{n-k}][Cp^{\alpha-1}(1-p)^{\beta-1}]}{\int_0^1 p^k(1-p)^{n-k} Cp^{\alpha-1}(1-p)^{\beta-1} dp} = \\ &= \frac{p^{\alpha+k-1}(1-p)^{\beta+n-k-1}}{\int_0^1 p^{\alpha+k-1}(1-p)^{\beta+n-k-1} dp} = \\ &= \frac{1}{B(\alpha+k, \beta+n-k)} p^{\alpha+k-1}(1-p)^{\beta+n-k-1} \end{aligned} \quad (21)$$

Conjugated priors

- If the prior is beta with α and β , then the posterior will be beta with $\alpha + k$ and $\beta + n - k$
- The mode of the beta distribution is $\frac{\alpha-1}{\alpha+\beta-2}$
- If we want a point estimate, we can take the posterior mode

$$\frac{\alpha - 1 + k}{\alpha + \beta - 2 + n}$$

- Note that with $\alpha = \beta = 1$, the posterior mode is the ML estimate
- This is because in this case the prior is flat, i.e. there's effectively no prior information
- Similarly, when n and k are very large, the prior becomes negligible, as the information in the data will completely take over

Univariate normal mean I

- We want to estimate the mean of a normal distribution $\mathcal{N}(\mu, \sigma^2)$ μ , assuming σ^2 is known
 - This is basically a linear regression with known σ^2 and fixed $\beta = 0$
- The likelihood of a sample $y_1, \dots, y_T$ will be

$$f(y|\mu) = (2\pi\sigma^2)^{-T/2} e^{-\sum_{t=1}^T \frac{(y_t - \mu)^2}{2\sigma^2}} \quad (22)$$

- Since the likelihood is "normal" also as a function of μ , choose the prior as a $\mathcal{N}\left(\mu_0, \frac{\sigma^2}{\nu_0}\right)$ normal

$$f(\mu) = \sqrt{\frac{\nu_0}{2\pi\sigma^2}} e^{-\nu_0 \frac{(\mu - \mu_0)^2}{2\sigma^2}} \quad (23)$$

- The prior has two parameters: μ_0 is the prior mean, ν_0 is the prior *relative precision*

Univariate normal mean II

- The posterior

$$\begin{aligned}
 f(\mu|y) &= \frac{(2\pi\sigma^2)^{-T/2} e^{-\sum_{t=1}^T \frac{(y_t-\mu)^2}{2\sigma^2}} \sqrt{\frac{\nu_0}{2\pi\sigma^2}} e^{-\nu_0 \frac{(\mu-\mu_0)^2}{2\sigma^2}}}{\int_{-\infty}^{\infty} (2\pi\sigma^2)^{-T/2} e^{-\sum_{t=1}^T \frac{(y_t-\mu)^2}{2\sigma^2}} \sqrt{\frac{\nu_0}{2\pi\sigma^2}} e^{-\nu_0 \frac{(\mu-\mu_0)^2}{2\sigma^2}} d\mu} = \\
 &= \frac{e^{-\frac{1}{2\sigma^2} (\sum_{t=1}^T (y_t-\mu)^2 + \nu_0 (\mu-\mu_0)^2)}}{\int_{-\infty}^{\infty} e^{-\frac{1}{2\sigma^2} (\sum_{t=1}^T (y_t-\mu)^2 + \nu_0 (\mu-\mu_0)^2)} d\mu} \quad (24)
 \end{aligned}$$

- From this we expect the posterior to be normal as well, since the exponent is quadratic in μ

Univariate normal mean III

$$\begin{aligned}
 \sum_{t=1}^T (y_t - \mu)^2 + \nu_0 (\mu - \mu_0)^2 &= \sum_{t=1}^T (y_t^2 - 2y_t\mu + \mu^2) + \nu_0 (\mu - \mu_0)^2 = \\
 &= (T + \nu_0)\mu^2 - 2 \left(\sum_{t=1}^T y_t + \nu_0 \mu_0 \right) \mu + \left(\sum_{t=1}^T y_t^2 + \nu_0 \mu_0^2 \right) = \\
 &= (T + \nu_0) \left(\mu^2 - 2 \frac{T\bar{y} + \nu_0 \mu_0}{T + \nu_0} \mu \right) + TS^2 + T\bar{y}^2 + \nu_0 \mu_0^2 = \tag{25} \\
 &= (T + \nu_0) \left(\mu - \frac{T\bar{y} + \nu_0 \mu_0}{T + \nu_0} \right)^2 + TS^2 + T\bar{y}^2 + \nu_0 \mu_0^2 - \frac{(T\bar{y} + \nu_0 \mu_0)^2}{T + \nu_0} = \\
 &= (T + \nu_0) \left(\mu - \frac{T\bar{y} + \nu_0 \mu_0}{T + \nu_0} \right)^2 + TS^2 + \frac{T\nu_0 \bar{y}^2 + T\nu_0 \mu_0^2 - 2T\nu_0 \bar{y} \mu_0}{T + \nu_0} = \\
 &= (T + \nu_0) \left(\mu - \frac{T\bar{y} + \nu_0 \mu_0}{T + \nu_0} \right)^2 + TS^2 + \frac{T\nu_0}{T + \nu_0} (\bar{y} - \mu_0)^2
 \end{aligned}$$

where $\bar{y} = \frac{1}{T} \sum y_t$, and $s^2 = \frac{1}{T} \sum (y_t - \bar{y})^2$

Univariate normal mean IV

- Let

$$\nu_1 = T + \nu_0, \quad \mu_1 = \frac{T\bar{y} + \nu_0\mu_0}{T + \nu_0}, \quad K = Ts^2 + \frac{T\nu_0}{T + \nu_0}(\bar{y} - \mu_0)^2 \quad (26)$$

- Then the posterior is

$$f(\mu|y) = \frac{e^{-\frac{1}{2\sigma^2}(\nu_1(\mu-\mu_1)^2+K)}}{\int_{-\infty}^{\infty} e^{-\frac{1}{2\sigma^2}(\nu_1(\mu-\mu_1)^2+K)} d\mu} = \frac{e^{-\frac{\nu_1}{2\sigma^2}(\mu-\mu_1)^2}}{\int_{-\infty}^{\infty} e^{-\frac{\nu_1}{2\sigma^2}(\mu-\mu_1)^2} d\mu} \quad (27)$$

which is a $\mathcal{N}\left(\mu_1, \frac{\sigma^2}{\nu_1}\right)$ distribution

Univariate normal mean V

- With known variance, the conjugate prior to a normal is a normal
- The posterior mean is a weighted average of the prior mean and the ML estimate, the weights being the relative precisions (the standard deviation of the ML estimate is σ^2/T , so the relative precision is T)
- The relative precision of the posterior is the sum of the prior and ML relative precisions

Univariate normal variance I

- The likelihood is

$$f(y|\sigma^2) = (2\pi)^{-T/2} (\sigma^2)^{-T/2} e^{-\frac{1}{2}Ts^2/\sigma^2} \quad (28)$$

where $s^2 = \frac{1}{T} \sum (y_i - \mu)^2$, the ML estimate of σ^2

- Let's define the prior so that it's similar in form

$$f(x) = C(\alpha, \beta) x^{-(\alpha+1)} e^{-\frac{\beta}{x}}, \quad 0 < \alpha, \beta, x \quad (29)$$

this is called an *inverse gamma* distribution

Univariate normal variance II

- With this, the posterior will be

$$\begin{aligned}
 f(\sigma^2|y) &= \frac{(2\pi)^{-T/2} C(\alpha_0, \beta_0) (\sigma^2)^{-T/2-\alpha_0-1} e^{-\frac{1}{\sigma^2}(\beta_0 + \frac{1}{2}Ts^2)}}{\int_0^\infty (2\pi)^{-T/2} C(\alpha_0, \beta_0) (\sigma^2)^{-T/2-\alpha_0-1} e^{-\frac{1}{\sigma^2}(\beta_0 + \frac{1}{2}Ts^2)} d\sigma^2} = \\
 &= \frac{(\sigma^2)^{-(\alpha_0+T/2)-1} e^{-\frac{1}{\sigma^2}(\beta_0 + \frac{1}{2}Ts^2)}}{\int_0^\infty (\sigma^2)^{-(\alpha_0+T/2)-1} e^{-\frac{1}{\sigma^2}(\beta_0 + \frac{1}{2}Ts^2)} d\sigma^2}
 \end{aligned} \tag{30}$$

- This is also inverse gamma with $\alpha_1 = \alpha_0 + \frac{T}{2}$ és $\beta_1 = \beta_0 + \frac{1}{2}Ts^2$
- The ML estimate of σ^2 has a χ^2 distribution, which is a special case of inverse gamma

$$(\chi^2)^{-1}(v) = \Gamma^{-1}(v/2, 1/2) \tag{31}$$

Univariate both parameters

- If we want to estimate both parameters, it is useful to define the prior as a product

$$f(\mu, \sigma^2) = f(\mu|\sigma^2)f(\sigma^2) \quad (32)$$

- This gives the joint distribution of (μ, σ^2) as the product of σ^2 distribution and the conditional distribution of μ
- We call this a normal - inverse gamma distribution

$$f(\mu, \sigma^2 | \mu_0, \nu_0, \alpha_0, \beta_0) = \sqrt{\frac{\nu_0}{2\pi\sigma^2}} C(\alpha_0, \beta_0) (\sigma^2)^{-(\alpha_0+1)} e^{-\frac{2\beta_0 + \nu_0(\mu - \mu_0)^2}{2\sigma^2}} \quad (33)$$

- The posterior will be

$$\begin{aligned} \mu_1 &= \frac{T\bar{y} + \nu_0\mu_0}{T + \nu_0}, \quad \nu_1 = \nu_0 + T \\ \alpha_1 &= \alpha_0 + \frac{T}{2}, \quad \beta_1 = \beta_0 + \frac{1}{2}Ts^2 + \frac{1}{2}\frac{T\nu_0}{T + \nu_0}(\bar{y} - \mu_0)^2 \end{aligned} \quad (34)$$

The Metropolis-Hastings algorithm I

- In most cases, the posterior density can not be determined analytically
- A common example is when we restrict $0 \leq b \leq 1$ with a beta prior
- Even though the posterior cannot be determined analytically, the basic identity still holds

$$f(\theta|y) \propto f(y|\theta)f(\theta) \quad (35)$$

- This means that we can always evaluate the posterior density up to a normalizing constant
- The Metropolis-Hastings (MH) sampler uses only this information

The Metropolis-Hastings algorithm II

- We need a so called proposal distribution $g(\theta)$, which is easy to sample from, and it has the same support as the target distribution $f(\theta)$
 - The set of possible values is the same for the two distributions
- We set an initial value $\theta^{(0)}$, and then in each step i
 - We generate a random number θ^c from g ; this will be called the candidate value
 - Then evaluate the acceptance ratio

$$\alpha = \frac{f(\theta^c)}{f(\theta^{(i-1)})} \quad (36)$$

- Accept the candidate value with probability α

$$\theta^{(i)} = \begin{cases} \theta^c, & \text{with probability } \alpha \\ \theta^{(i-1)}, & \text{with probability } 1 - \alpha \end{cases} \quad (37)$$

The Metropolis-Hastings algorithm III

- Since the acceptance ratio in (36) depends only on the ratio of the target density at two points, we do not need to know the normalizing constant
- If the candidate is "more likely" than the previously accepted value, we accept the candidate
- But even if it's "less likely", it is still accepted with some probability, which ensures that less likely regions of the target distribution are sampled
- The proposal density can actually be made to depend on the previously accepted value
 - It is common to set g to a normal with mean $\theta^{(i-1)}$
- The resulting sample can be highly autocorrelated, especially when the acceptance ratio is small
 - We should only retain values which are sufficiently apart

Adaptive proposal distribution I

- Another possibility is the change the proposal density to (hopefully) get close to a desired acceptance ratio
 - Values between 0.1 - 0.6 are considered fine in the literature
- A too low acceptance ratio results in simulation with long flat sections
- A too large acceptance ratio results in a highly correlated sample
- Turn, low/high acceptance ratio is the result of too large/small proposal variance
 - As indicated by a large/small α

Adaptive proposal distribution II

- A reasonable update formula might be

$$\sigma_{\text{new}} = \sigma_{\text{old}} e^{\lambda n^{-\gamma}(\alpha - \bar{\alpha})} \quad (38)$$

where $\bar{\alpha}$ is the target acceptance ratio, n is the current iteration, and $0 \leq \lambda$ and $0 \leq \gamma$ are parameters governing the adaptation speed

- We would like adaptation to be more aggressive at the beginning of the simulation, therefore the adaptation factor is decreasing in n
- We use mainly λ to set the adaptation speed: larger λ implies more dynamic adaptation